A Quarter of the Budget Is Spent Twice: Duplicated Evidence in Budgeted Context Packing and a Simple Fix

Anonymous ACL submission

Abstract

When a long document must be answered from under a hard token budget, the standard recipe ranks chunks with a retriever, pads each with neighbouring text, and concatenates the top ones until the budget runs out. Research has focused on the ranking step; we look at the packing step and find that it is surprisingly wasteful. On LongBench, standard dense chunk packing spends 23% of a 1,024-token budget, and 29% of a 2,048-token budget, on text that already appears elsewhere in the same context, because neighbouring top-ranked chunks are padded into each other; almost every packed piece also ends mid-sentence. We present SENTPACK, a training-free packer that removes this waste behind any ranker: it extends each selected span to sentence boundaries, merges overlapping extensions so that no text is paid for twice, admits evidence under the exact token count of the downstream tokenizer with backfilling, and emits it in document order. Duplication drops to 1.4%, and the recovered budget raises answer coverage by 3–4 points. We freeze every design choice on a development split and run the LongBench test set (1,150 single- and multi-document QA instances) once, with three instruction models (Qwen2.5-7B, Llama-3.1-8B, Qwen2.5-14B) and three budgets (512, 1,024, 2,048 tokens). SENTPACK outperforms dense BGE chunk packing in all nine model–budget settings, by 1.9–3.7 F1 when pooled over models ($p \leq 0.001$ at every budget), and outperforms a structured lexical pair packer by 2.5–3.1 F1. The gain does not shrink with better rankers: holding the ranking fixed, SENTPACK improves chunk packing by 2.8, 3.8 and 2.0 F1 with BGE-small, BGE-large and a cross-encoder reranker, and it outperforms LlamaIndex's sentence-window and auto-merging retrievers and LongLLMLingua. At 2,048 tokens, SENTPACK is statistically indistinguishable from reading the full context on all three models, and it exceeds LLMLingua-2 at the same budget by 15.3 F1.

1 Introduction

Large language models (LLMs) are routinely asked to answer questions about long documents and document collections. Even when the context window is long enough, there are reasons to send the model less text: prefill cost and latency grow with prompt length, many deployed models have short windows, and long inputs dilute the few passages that matter (Liu et al., 2024a). In practice a long context is therefore often reduced to a *hard* budget of a few hundred to a few thousand tokens, measured in the tokenizer of the model that will read it.

Two families of methods do this. *Prompt compressors* (Li et al., 2023; Jiang et al., 2023, 2024; Pan et al., 2024) delete tokens or phrases judged uninformative. *Retrieval-style packers* split the context into chunks or spans, rank them with BM25 or a dense encoder (Robertson and Zaragoza, 2009; Karpukhin et al., 2020; Xiao et al., 2023b), and concatenate the top-ranked units, each padded with a fixed window of neighbouring tokens, until the budget is exhausted. Research effort has concentrated on the ranking step; the packing step is treated as bookkeeping.

We start from a simple question: *where does the budget go?* Measuring the contexts produced by standard dense chunk packing on LongBench (Section 3), we find that relevant chunks cluster—the passage that answers a question tends to contain several top-ranked chunks—so their padded windows overlap, and the same sentences are rendered and paid for several times. At a 1,024-token budget, 23% of the packed text duplicates text elsewhere in the same context, and the share grows with the budget (15%, 23% and 29% at 512, 1,024 and 2,048 tokens). In addition, 94% of packed pieces end in the middle of a sentence, and greedy admission leaves part of the budget unused. Under a hard budget, every duplicated or broken token is a token not spent on evidence.

We then show that this waste can be removed without touching the ranker. SENTPACK is a packer that can be placed behind any span ranker (Figure 1). SENTPACK (i) extends each selected span to the nearest sentence boundaries within a small radius, (ii) merges extensions that touch or overlap into a single passage, (iii) admits spans in rank order under the exact downstream token count and *backfills*, i.e., keeps admitting later spans after one does not fit, and (iv) emits the merged passages verbatim in document order. None of these operations requires training, and all of them run on the CPU before the model is called.

We evaluate SENTPACK under a protocol designed to leave little room for doubt. Every design choice, including the ranker, was fixed on a development split built from LongBench-E with all LongBench test questions removed; the LongBench test set was then run once. All methods share the candidate spans, the prompt wrapper, the model, greedy decoding, the scorer, and a budget counted with the downstream tokenizer. We test three instruction models and three budgets and report paired bootstrap tests. Our findings are:

- **Standard packing wastes a quarter of the budget.** Dense chunk packing duplicates 15–30% of its context; SENTPACK reduces this to about 1.5% and raises answer coverage by 3–4 points at every budget (Section 3).
- **Removing the waste beats ranking-only improvements.** SENTPACK outperforms dense BGE chunk packing in all nine model–budget settings, by +3.68, +1.92 and +2.26 F1 at 512, 1,024 and 2,048 tokens (pooled over models, all $p \leq 0.001$), and outperforms a structured lexical pair packer by 2.5–3.1 F1 (Tables 2 and 3).
- **Better rankers do not remove the waste.** With BGE-large and with a cross-encoder reranker, swapping only the packer improves F1 by 3.8 and 2.0 ($p \leq 0.001$); SENTPACK with the reranker is the best compressed configuration and outperforms LlamaIndex's sentence-window and auto-merging retrievers, Provence and LongLLMLingua (Section 6.3). The direction carries over to book-length InfiniteBench contexts (Section 6.4).
- **The gain is retriever-agnostic.** Holding the BGE ranking fixed and changing only the packer improves F1 by 2.5–2.8 at every budget; with a purely lexical ranker SENTPACK still outperforms the pair packer that uses the same lexical signals (Section 6.2).
- **A 2,048-token SENTPACK context is as good as the full context** on all three models (paired differences $+0.56$, -0.35 and -0.10 F1, none significant), while sending about one sixth of the prompt.
- **Evidence-level packing dominates token deletion.** At the same exact budget, SENTPACK exceeds LLMLingua-2 by 15.3 F1.
- **The packer is robust.** Removing any one of its components still leaves a packer that beats BGE chunk packing; merging overlapping evidence is the component whose removal costs the most (Section 6.7).

2 Related Work

Prompt and context compression. Selective Context removes lexical units with low self-information (Li et al., 2023). LLMLingua and LongLLMLingua estimate token importance with a small LM, the latter with question-aware coarse-to-fine compression (Jiang et al., 2023, 2024); LLMLingua-2 distils keep/drop labels into a bidirectional encoder (Pan et al., 2024). CPC scores sentences against the question with a trained encoder (Liskavets et al., 2025), and RECOMP and CompAct train extractive or abstractive compressors (Xu et al., 2024; Yoon et al., 2024). SENTPACK is training-free, never edits text inside a kept passage, and is complementary to learned scorers: any of them can supply its ranking.

Retrieval-style context construction. Retrieval-augmented generation ranks passages with sparse (Robertson and Zaragoza, 2009) or dense (Karpukhin et al., 2020; Xiao et al., 2023b) retrievers and concatenates the top results (Lewis et al., 2020; Izacard and Grave, 2021); diversity-aware selection (Carbonell and Goldstein, 1998) and multi-hop retrieval (Xiong et al., 2021; Trivedi et al., 2023) refine *which* units are selected. We study the complementary question of how selected units are turned into a context under an exact budget, and show that this step alone moves F1 by 2–4 points.

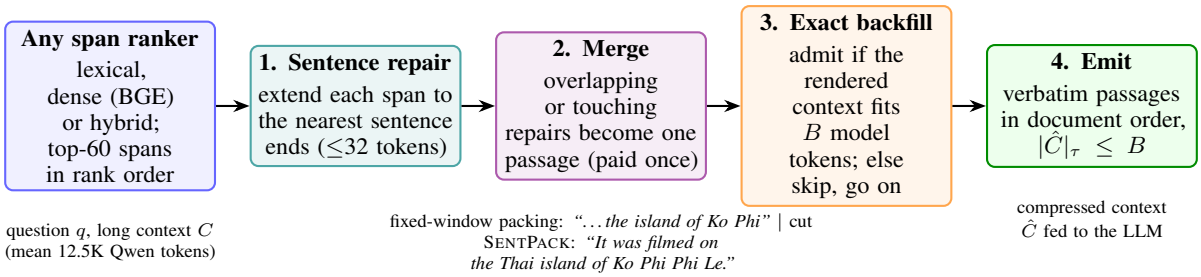


Figure 1: SENTPACK packs the spans returned by any ranker. Each span is extended to sentence boundaries, overlapping extensions are merged, spans are admitted in rank order under the exact downstream token budget with backfilling, and the result is emitted verbatim in document order. SENTPACK runs before prefill and does not modify the model.

B	Packer	Dup. ↓	Sent. end ↑	Tokens	Ans. cov. ↑
512	BGE chunks	14.7	6.7	473	30.4
	Pair packing	3.0	6.2	439	30.0
	SENTPACK	**1.1**	**70.6**	500	**33.9**
1,024	BGE chunks	22.8	6.3	985	39.0
	Pair packing	5.4	6.0	920	36.9
	SENTPACK	**1.4**	**68.1**	1,014	**41.9**
2,048	BGE chunks	29.1	6.2	2,010	45.0
	Pair packing	7.3	6.1	1,502	41.2
	SENTPACK	**1.6**	**62.8**	2,032	**49.0**

Table 1: Where the budget goes (LongBench test, 1,150 instances, Qwen tokenizer; Llama-3.1 is within one point on every column). Dup.: share of packed words that start an 8-word sequence already present earlier in the same context. Sent. end: share of packed pieces ending with sentence punctuation. Tokens: mean context length. Ans. cov.: share of instances whose normalised gold answer appears verbatim in the context.

Long-context evaluation and KV-cache compression. LongBench (Bai et al., 2023) evaluates QA over long inputs, and models under-use evidence in the middle of long contexts (Liu et al., 2024a). KV-cache methods evict, merge or quantise cached states during generation (Xiao et al., 2023a; Zhang et al., 2023; Li et al., 2024; Liu et al., 2024b). SENTPACK instead shortens the input before prefill, so the full context is never encoded.

3 Where Does the Budget Go?

We measure three kinds of waste in the contexts produced by standard packers on the LongBench test set (Section 5), before any model reads them (Table 1).

Duplication. We count a packed word as duplicated if the 8-word sequence it starts already appeared earlier in the same packed context. Dense chunk packing duplicates 14.7%, 22.8% and 29.1% of its text at 512, 1,024 and 2,048 tokens. The cause is structural: the top-ranked candidate spans for a question are rarely spread evenly over the document; they cluster in the passages that are actually relevant, and adjacent spans overlap by construction (64-token spans with stride 32). Padding each span by ±32 tokens then makes neighbouring windows overlap almost completely. The larger the budget, the more neighbours are admitted, and the larger the duplicated share. A structured pair packer that penalises redundancy at selection time reduces duplication but does not remove it (3–7%), because overlap is created by the padding, not by the selection.

Broken sentences. Only about 6% of the pieces produced by fixed-window packing end with sentence punctuation; the remaining 94% stop mid-sentence, often inside the phrase that answers the question.

Unused budget. Greedy admission that stops or skips on long units leaves part of the budget unfilled: on average 39 of 1,024 tokens for chunk packing and 104 for pair packing.

Consequence. The three effects combine into lower answer coverage: at every budget, the gold answer is less often present in the packed context than it could be. The packer we describe next removes all three: duplication drops to 1.1–1.6%, two thirds of pieces end on a sentence boundary, the budget is filled to within 10 tokens, and answer coverage rises by 2.9–4.0 points over dense chunk packing with the same budget.

4 Method

4.1 Problem Setting

Given a question q and a long context C, we construct $\hat{C}$ with $|\hat{C}|_\tau \leq B$, where $|\cdot|_\tau$ counts tokens with the tokenizer τ of the downstream model. A fixed prompt wrapper keeps the task instruction and question unchanged; only the context slot is replaced by $\hat{C}$. Because the budget is defined in downstream tokens, every method is held to exactly the same constraint for each model.

4.2 Candidate Spans

The context is split into contiguous pages of 256 regex tokens (words or single punctuation marks), and each page into candidate spans of 64 regex tokens with stride 32; spans never cross pages. Regex tokens only define candidate units; all budget accounting uses τ. All span-based methods in our comparison use the same candidate spans.

4.3 Rankers

SENTPACK accepts any function that scores spans against q. We use three, none of which sees gold answers. The *lexical* ranker fuses z-normalised BM25 (k_1=1.2, b=0.75) and TF-IDF scores computed over the spans of the instance. The *dense* ranker is the cosine similarity between the question (with BGE's query instruction) and the span under `bge-small-en-v1.5` (Xiao et al., 2023b). The *hybrid* ranker adds the z-normalised lexical and dense scores. The top 60 spans form the priority list P, ordered by score with ties broken by position. Our main configuration uses the hybrid ranker, chosen on the development split; Section 6.2 shows the gains do not depend on this choice.

4.4 The Packer

Sentence repair. Fixed-size spans start and end at arbitrary positions. Rather than padding each span by a fixed window, SENTPACK extends its left end backwards to the first token after a sentence terminator (`.`, `?`, `!`) and its right end forwards to the next terminator, each by at most $r = 32$ regex tokens and within the page. A repaired span is thus never longer than a fixed $\pm r$ window, and usually shorter, but it starts and ends on sentence boundaries whenever one is within reach.

Merging. Spans that are close in the document produce repaired ranges that overlap or touch. SENTPACK sorts the repaired ranges of the current selection by page and position and merges every pair that overlaps into one passage. Overlapping evidence is therefore rendered, and paid for, once.

Exact admission with backfill. Spans are considered in priority order. A span is admitted if the context rendered from the current selection plus the span (repaired, merged and joined after the prefix "Question: q / Selected evidence:") has at most B tokens under τ. Because merging can make a new span nearly free, the cost of a span depends on what is already selected, so it is measured on the rendered context rather than estimated. If a span does not fit, SENTPACK skips it and continues with the next one, so short spans further down the list can still use the remaining budget.

Document-order emission. The merged passages are emitted verbatim in document order, separated by blank lines. If re-joining changes the token count, the context is clipped to B, so $|\hat{C}|_\tau \leq B$ always holds. Algorithm 1 (Appendix A) gives the procedure.

Development protocol. All choices—the ranker, sentence rather than fixed repair, r, the list length, merging, backfilling and emission order—were made on a development split of 584 instances: the HotpotQA and 2WikiMultihopQA portions of LongBench-E (Bai et al., 2023) after removing every question that also appears in the LongBench test set. The development experiments used a 4-bit Qwen2.5-7B model on a separate GPU. The configuration was then frozen and the test set was run once with every model and budget reported here; no test result was used to change the method.

5 Experimental Setup

Data. All English single- and multi-document QA tasks of LongBench (Bai et al., 2023): Qasper (Dasigi et al., 2021) (200), MultiFieldQA-en (150), HotpotQA (Yang et al., 2018) (200), 2WikiMultihopQA (Ho et al., 2020) (200), MuSiQue (Trivedi et al., 2022) (200) and NarrativeQA (Kočiský et al., 2018) (200), 1,150 instances. We report LongBench's F1 and the macro average over the six tasks; "MH" averages the three multi-hop tasks.

Models and decoding. Qwen2.5-7B-Instruct, Qwen2.5-14B-Instruct (Qwen Team, 2024) and Llama-3.1-8B-Instruct (Grattafiori et al., 2024), in bfloat16 with vLLM, LongBench's task prompts inside each model's chat template, greedy decod-

ing and LongBench's per-task output lengths. The 14B model reads the packed contexts built with the Qwen tokenizer, which it shares with the 7B model.

Budgets. $B \in \{512, 1{,}024, 2{,}048\}$ downstream tokens for the context slot. The mean full prompt has 12.5K Qwen tokens, so these budgets keep roughly 4%, 8% and 16% of the context.

Baselines. All baselines fill the same slot under the same exact budget and tokenizer. *BGE chunks*: the 64 spans most similar to the question under BGE, each padded by ± 32 tokens, admitted in similarity order while they fit and emitted in that order; this is the standard dense packing recipe. *Pair packing*: a structured lexical packer that selects query–support span pairs with the lexical ranker, a lexical bridge score and a redundancy penalty, repairs each span with a fixed ± 32 window, admits pairs until the first one that does not fit and emits them in document order. It represents structured, multi-hop-aware selection with conventional packing. *LLMLingua-2* (Pan et al., 2024) (`xlm-roberta-large-meetingbank`), asked to compress the context to the budget; its output is placed after the same prefix and truncated to B downstream tokens if needed. *Full context*: the whole context, middle-truncated to 30K tokens as in LongBench (a 32K window for all models), as a reference. Section 6.3 adds stronger rankers (BGE-large; `bge-reranker-v2-m3` over the top 100 BGE spans) and published systems: LlamaIndex's sentence-window and auto-merging retrievers (Liu, 2022) (official node parsers, default window 3 and hierarchy 2048/512/128, merge threshold 0.5, BGE-small retrieval), sentence-aligned chunks of up to 64 words ranked by BGE, Provence (Chirkova et al., 2025) applied to the top 60 BGE chunks, and LongLLMLingua (Jiang et al., 2024) with its recommended question-aware settings. For LongLLMLingua we use Llama-3.1-8B-Instruct as the compressor LM; it failed on 26 of 1,150 instances (2.3%), for which we fall back to the head of the context. All of these are packed into the same exact budget; units are admitted in their own rank order while they fit. Appendix E compares the pair packer with truncation, Selective Context and BM25/TF-IDF chunk packing in an earlier environment.

Statistics. Comparisons are paired on instances. We use a task-stratified paired bootstrap (Efron and Tibshirani, 1993) with 10,000 resamples and seed 20260701; the statistic is the macro average of per-task mean differences. Pooled results resample within the eighteen model–task strata. We report 95% percentile intervals and two-sided p-values.

6 Results

6.1 SentPack versus standard packing

Table 2 gives macro F1 for every model and budget, and Table 3 the pooled paired tests. SentPack outperforms BGE chunk packing in all nine model–budget settings. Pooled over models, the gain is +3.68 F1 at 512 tokens, +1.92 at 1,024 and +2.26 at 2,048, and every interval excludes zero. Per model, eight of the nine differences are significant at $p < 0.05$ (the ninth, Qwen2.5-7B at 1,024 tokens, is +1.79 with $p = 0.06$; Appendix D). The gain is largest at the smallest budget, where every wasted token is most costly.

SentPack also outperforms the structured pair packer by 2.5–3.1 F1 at every budget ($p < 0.001$), in all nine settings. The pair packer was designed for multi-hop questions and selects evidence in pairs, yet a packer that ranks spans one at a time but packs them carefully is better on the multi-hop tasks as well: at 1,024 tokens SentPack reaches 46.3, 43.9 and 50.3 MH F1 on the three models, against 42.4, 39.6 and 47.0 for pair packing and 42.7, 41.3 and 48.0 for BGE chunks (Appendix C).

6.2 The packer, not the ranker, carries the gain

The most direct test of the packer is to keep the ranking fixed. BGE chunk packing and SentPack with the BGE ranker use the same BGE similarities over the same spans and differ only in how the top spans are turned into a context. Changing the packer improves F1 by +2.81, +2.48 and +2.57 at the three budgets (Table 3), and per model it is significant in all nine settings (Appendix D).

Conversely, the choice of ranker inside SentPack matters little. The hybrid ranker selected on the development split and the BGE ranker are within 0.6 F1 of each other on average and their difference is not significant (-0.56, $[-1.49, 0.40]$ at 1,024 tokens); each is best in some settings. With a purely lexical ranker, SentPack outperforms the pair packer, which uses the same lexical signals, by 1.54 F1 [0.68, 2.43] ($p < 0.001$) at 1,024 tokens. In short, the improvement comes from the packing operations and transfers across lexical, dense and hybrid rankers.

Method	Qwen2.5-7B 512	Qwen2.5-7B 1,024	Qwen2.5-7B 2,048	Llama-3.1-8B 512	Llama-3.1-8B 1,024	Llama-3.1-8B 2,048	Qwen2.5-14B 512	Qwen2.5-14B 1,024	Qwen2.5-14B 2,048	Avg.
Full context (reference)		43.04			43.13			47.59		
LLMLingua-2	–	24.60	–	–	28.43	–	–	27.10	–	–
Pair packing	33.88	39.11	40.68	34.05	37.79	39.60	36.73	41.69	44.23	38.64
BGE chunks	33.05	38.85	40.98	33.27	38.99	40.85	36.27	42.34	45.29	38.88
SENTPACK (BGE ranker)	35.64	**40.76**	43.14	36.16	**41.67**	**43.58**	39.21	**45.18**	**48.10**	**41.49**
SENTPACK (hybrid ranker, main)	**36.74**	40.64	**43.60**	**36.44**	40.98	42.78	**40.44**	44.32	47.50	**41.49**

Table 2: Macro F1 on the 1,150 LongBench QA instances under exact budgets of 512, 1,024 and 2,048 downstream tokens. The full-context reference reads up to 30K tokens. Bold marks the best compressed result in each column. Per-task scores are in Appendix C.

SENTPACK minus	512	1,024	2,048
BGE chunks	**+3.68** [2.53, 4.82]	**+1.92** [0.87, 2.97]	**+2.26** [1.24, 3.29]
Pair packing	**+2.99** [2.04, 3.94]	**+2.45** [1.52, 3.39]	**+3.12** [2.18, 4.08]
Same BGE ranking, different packer:			
SENTPACK-BGE − BGE chunks	**+2.81** [1.88, 3.76]	**+2.48** [1.50, 3.45]	**+2.57** [1.58, 3.57]

Table 3: Paired F1 differences pooled over the three models (95% bootstrap intervals). All intervals exclude zero ($p \leq 0.001$). Per-model tests are in Appendix D.

Ranker	Packing / system	Qwen-14B	Llama-8B	Δ (pooled)
BGE-small	chunks	42.34	38.99	
	SENTPACK	45.18	41.67	**+2.76*****
BGE-large	chunks	42.55	40.12	
	SENTPACK	46.23	43.98	**+3.77*****
BGE + bge-reranker-v2-m3	chunks	44.37	42.21	
	SENTPACK	**46.45**	**44.12**	**+2.00*****
LlamaIndex sentence window		39.99	36.96	
LlamaIndex auto-merging		41.50	39.13	
Sentence chunks + BGE		44.24	39.98	
Provence (BGE + pruning)		44.27	40.69	
LongLLMLingua		40.66	39.67	
LLMLingua-2		27.10	28.43	
Full context (30K)		47.59	43.13	

Table 4: Stronger rankers and external context-construction systems (LongBench, 1,024 tokens, macro F1). Δ: SENTPACK minus chunk packing *with the same ranking*, pooled over the two models (***: $p \leq 0.001$; significant for each model separately as well).

6.3 Stronger rankers and external systems

A natural question is whether the waste we identify matters only for weak rankers. Table 4 repeats the packer swap with two stronger rankers on Qwen2.5-14B and Llama-3.1-8B. With BGE-large, SENTPACK improves on chunk packing by 3.77 F1 [2.67, 4.88]; with a cross-encoder reranker (`bge-reranker-v2-m3` over the top 100 BGE spans) by 2.00 [0.85, 3.12]. Both gains are significant on each model separately. A better ranker therefore does not remove the packing waste: the gain from better packing is at least as large as the gain from replacing BGE-small with BGE-large or with a reranker, and the two add up. SENTPACK with the reranker is the best compressed configuration on both models, and on Llama-3.1-8B it scores above the 30K-token full context (44.12 vs. 43.13).

We also compare with published context-construction systems under the same budget and wrapper. LlamaIndex's sentence-window retrieval (Liu, 2022), which retrieves single sentences and replaces each with its ±3-sentence window, trails the frozen SENTPACK configuration by 4.18 F1 [2.76, 5.65]; its windows of neighbouring top sentences overlap heavily, which is exactly the duplication of Section 3. LlamaIndex's auto-merging retriever, which replaces retrieved leaf chunks by their parent when most of the parent's children are retrieved, trails it by 2.33 [0.94, 3.79]: merging at fixed hierarchy levels removes some duplication but admits whole parents that are mostly irrelevant. LongLLMLingua (Jiang et al., 2024) trails it by 2.48 [1.02, 3.98]. Provence (Chirkova et al., 2025), a trained pruner that removes irrelevant sentences from each retrieved chunk, is close to the frozen configuration (+0.17, n.s.) and is clearly below SENTPACK given a reranker (+2.80 [1.49, 4.16]); SENTPACK needs no trained pruner. Finally, packing non-overlapping sentence-aligned chunks with BGE, which avoids duplication by construction, recovers part of the gain, as our diagnosis predicts; SENTPACK with the same BGE ranker remains 1.31 F1 higher [−0.03, 2.67].

6.4 A second benchmark: book-length contexts

To test whether the findings transfer beyond LongBench, we run the frozen configuration on InfiniteBench's English novel QA and multiple-choice tasks (Zhang et al., 2024), whose contexts

Method	Qwen2.5-14B 1,024	Qwen2.5-14B 2,048	Llama-3.1-8B 1,024	Llama-3.1-8B 2,048
Pair packing	33.71	34.61	32.80	36.55
BGE chunks	36.58	37.44	36.81	40.15
Sentence chunks + BGE	36.36	37.69	36.09	39.99
SENTPACK (hybrid, frozen)	**38.03**	**38.45**	**38.36**	40.98
SENTPACK (BGE)	36.76	37.63	36.56	**41.49**
Full context (30K)	–		38.16	

Table 5: InfiniteBench En.QA and En.MC (Zhang et al., 2024) (580 questions over novels of 150K+ tokens), mean of QA F1 and MC accuracy. No setting was tuned on this benchmark.

are about ten times longer (Table 5). No parameter was changed. The frozen SENTPACK is the best method in three of the four model–budget settings and outperforms the pair packer by 4.54 F1 [3.02, 6.02] pooled ($p < 0.001$). Its advantage over dense chunk packing points in the same direction as on LongBench (+1.21 pooled, between +0.84 and +1.56 per setting) but, with 580 questions, is not significant. With only 1,024 tokens, SENTPACK matches Llama-3.1-8B reading 30K tokens of the book (38.36 vs. 38.16).

6.5 How close to the full context?

At 2,048 tokens, SENTPACK is statistically indistinguishable from reading the full context on all three models: the paired differences are $+0.56$ $[-1.46, 2.56]$ for Qwen2.5-7B, -0.35 $[-2.39, 1.65]$ for Llama-3.1-8B and -0.10 $[-2.04, 1.82]$ for Qwen2.5-14B. The packed context is about one sixth of the mean prompt (2,048 vs. 12.5K tokens). On the multi-hop tasks it exceeds the full context on all three models (49.2 vs. 44.4, 45.8 vs. 43.7 and 56.0 vs. 52.6 MH F1), consistent with the observation that long inputs dilute the evidence that matters (Liu et al., 2024a). At 1,024 tokens, SENTPACK recovers 93–95% of full-context F1.

6.6 Evidence-level vs. token-level compression

Under the same exact 1,024-token budget, LLMLingua-2 reaches 24.6–28.4 F1, and SENTPACK exceeds it by 15.27 F1 [13.90, 16.64] pooled over models. At roughly $12\times$ compression a token-level compressor must remove most of every sentence, including the names, numbers and function words that extractive QA needs to copy, whereas SENTPACK spends the budget on a few complete passages. Appendix E shows the same ordering for Selective Context and LongLLMLingua.

Variant (1,024 tokens)	vs. BGE chunks	Full − variant
SENTPACK (all components)	$+1.92^*$	–
without merging	$+1.14^*$	$+0.78^*$ [0.03, 1.52]
without backfill	$+1.63^*$	+0.29 [−0.13, 0.71]
fixed ±32 repair	$+1.92^*$	+0.00 [−0.75, 0.73]
rank-order emission	$+2.14^*$	−0.22 [−0.97, 0.53]

Table 6: Component analysis, pooled over the three models (F1, paired bootstrap; *: 95% interval excludes zero). Each variant switches off one operation and keeps the hybrid ranker and the exact budget.

6.7 What each packing operation contributes

Table 6 switches off one packing operation at a time. Two observations stand out. First, the packer is robust: every single-component variant still outperforms BGE chunk packing significantly, so no single operation is a point of failure. Second, merging overlapping evidence is the operation whose removal costs the most (+0.78 F1, $p = 0.04$). Without merging, two selected spans from the same passage are rendered as two overlapping copies, and the duplicated text displaces other evidence; with merging, a span adjacent to already selected evidence costs only its new tokens, which also makes exact, rendered-context admission necessary. The remaining operations are interchangeable in the sense that one can compensate for another: sentence repair and fixed-window repair perform the same once merging and backfilling are in place, and so do document-order and rank-order emission. On the development split, sentence repair improved F1 by 2.5 over fixed repair with a lexical ranker (Appendix B); we keep it because it never lengthens a span beyond the fixed window and yields readable passages.

6.8 Cost

SENTPACK runs on a single CPU core in 432 ms per instance on average (median 415 ms, 95th percentile 698 ms, 1,024 tokens, 300 random test instances), compared with 614 ms for the pair packer and 280 ms for BGE chunk packing; the BGE encoder pass is shared by both dense methods. Every packed context respects the exact budget (maximum 1,024 tokens), and SENTPACK uses the budget more fully than chunk packing (1,014 vs. 987 tokens on average) because backfilling closes the gaps left by spans that do not fit.

7 Discussion

Why does packing matter so much? Under an exact budget, every token spent on a broken sen-

tence, a duplicated window or an unfilled gap is a token not spent on evidence. Standard packing pays all three costs. Sentence repair turns spans into self-contained statements, merging stops the same text from being bought twice, backfilling uses the remainder of the budget, and document order keeps neighbouring passages in their natural sequence. Each effect is small, and the operations partly substitute for one another, but together they are worth 2–4 F1, as much as replacing a lexical ranker with a dense one.

Implications for compression research. Most work on context compression compares rankers or learned scorers with a fixed, simple packer. Our results suggest that such comparisons should hold the packer fixed *and* strong, because a better packer gives every ranker a similar boost. SENTPACK needs only the downstream tokenizer and adds no model calls, so it can be adopted as the default packer behind any retriever or learned scorer.

Deployment. SENTPACK is training-free, runs before the model is called and outputs verbatim source text in document order, so the compressed context can be shown to users and traced back to its sources.

8 Conclusion

We studied long-context question answering under a hard budget measured in the downstream model's own tokens and showed that the step that turns ranked spans into a context is a major, understudied lever. SENTPACK—sentence repair, merging of overlapping evidence, exact admission with backfill and document-order emission—improves over dense chunk packing in all nine model–budget settings we test, by 1.9–3.7 F1 pooled over models, transfers across lexical, dense and hybrid rankers, and at 2,048 tokens matches the full context on three LLMs. We hope this encourages treating packing as a first-class component of context compression.

Limitations

SENTPACK shortens the input before prefill and is not a KV-cache method. Our experiments cover English extractive QA on LongBench with three instruction models of 7–14B parameters and budgets of 512–2,048 tokens; other languages, generation tasks such as summarisation, and much larger models may behave differently. The development split covers two multi-hop tasks, and the method was frozen before testing; a broader development set might lead to a different choice of ranker, although our results suggest the ranker matters less than the packer. Our component analysis shows that the packing operations partly substitute for one another, so their individual contributions are smaller than their joint effect. The dense and hybrid rankers require a small encoder at inference time.

Ethics Statement

Context compression can remove information a user would consider important, which can lead to incomplete or incorrect answers; in high-stakes settings, compressed contexts should be shown to the user or checked against the source. SENTPACK keeps all admitted text verbatim, which makes such checks straightforward. All datasets are public research benchmarks used under their licences.

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A Implementation Details

Algorithm 1 SENTPACK

Require: question q, context C, budget B, tokenizer τ, ranker f
split C into 256-token pages and 64-token spans (stride 32)
$P \leftarrow$ top 60 spans by $f(q, s)$
$\mathcal{S} \leftarrow \emptyset$
for $s \in P$ in order **do**
  $R \leftarrow$ sentence-repaired ranges of $\mathcal{S} \cup \{s\}$ ($\leq$32 tokens per side, within page)
  merge overlapping ranges of R; render in document order after the prefix
  if $|\text{render}|_\tau \leq B$ **then** $\mathcal{S} \leftarrow \mathcal{S} \cup \{s\}$
  end if ▷ otherwise skip s and continue (backfill)
end for
return render of $\mathcal{S}$, clipped to B tokens if needed

Inference used vLLM 0.7.3 with bfloat16 weights on NVIDIA A100-40GB GPUs (tensor parallelism of two for the 14B full-context run), with a 4,096-token model window for compressed contexts and 32,768 tokens for the full-context reference. Code, packed contexts and per-instance predictions for every number in this paper will be released.

B Development Results

Table 8 reports macro F1 on the development split (HotpotQA-E and 2WikiMultihopQA-E without test questions, 584 instances, 1,024 tokens, 4-bit Qwen2.5-7B-Instruct). These results determined the frozen configuration; the hybrid ranker with sentence repair was the best development configuration, +5.26 F1 [2.34, 8.23] above BGE chunk packing.

Item	Value
Pages / spans	256 regex tokens / 64, stride 32
Lexical ranker	$0.5\,z(\text{BM25}) + 0.5\,z(\text{TFIDF})$; k_1=1.2, b=0.75; sublinear TF, smoothed IDF
Dense ranker	`bge-small-en-v1.5`, CLS pooling, ℓ_2-normalised, query instruction
Hybrid ranker	$z(\text{lexical}) + z(\text{dense})$
Priority list	top 60 spans
Sentence terminators	. ? !
Repair radius	32 regex tokens per side, within page
Admission	rendered context under τ, backfill
Emission	document order, blank-line separated

Table 7: Implementation constants (all frozen before testing).

Ranker (all with merge + backfill)	Fixed $\pm$32	Sentence
Lexical	45.37	47.90
BGE	48.88	50.12
Hybrid	48.17	**51.19**
BGE chunks	45.93	
Pair packing	45.03	

Table 8: Development macro F1 (1,024 tokens).

C Per-Task Results

Table 9 lists per-task F1 at 1,024 tokens. Multi-hop (MH) F1 of SENTPACK (hybrid) at 512 / 2,048 tokens is 39.47 / 49.20 (Qwen2.5-7B), 37.02 / 45.78 (Llama-3.1-8B) and 44.18 / 56.01 (Qwen2.5-14B), against 34.84 / 46.37, 33.52 / 44.14 and 40.02 / 52.57 for BGE chunks.

D Per-Model Paired Tests

E Additional Baselines

In an earlier run with the same candidate spans, prompt wrapper, scorer and exact 1,024-token budget, but with Qwen2.5-7B-Instruct and Meta-Llama-3-8B-Instruct in an earlier software environment, the pair packer of Table 2 outperformed head and middle truncation by 13.1 and 13.6 F1, Selective Context by 18.2, LLMLingua-2 by 11.3, LongLLMLingua by 6.6, and BM25 and TF-IDF

Model	Method	Qasper	MF-en	HotpotQA	2Wiki	MuSiQue	NarrQA	MH	Avg.
Qwen2.5-7B	Full context	44.86	51.41	58.07	45.61	29.36	28.94	44.35	43.04
	LLMLingua-2	33.23	31.00	32.80	29.19	13.19	8.16	25.06	24.60
	Pair packing	40.40	50.08	54.05	46.25	26.83	17.05	42.38	39.11
	BGE chunks	38.90	47.90	51.61	45.88	30.72	18.08	42.74	38.85
	SENTPACK (BGE)	42.03	46.40	57.00	52.41	29.89	16.82	46.43	40.76
	SENTPACK (hybrid)	38.73	49.31	57.10	48.32	33.39	16.97	46.27	40.64
Llama-3.1-8B	Full context	45.28	54.80	55.79	44.14	31.26	27.52	43.73	43.13
	LLMLingua-2	34.31	33.35	37.35	31.53	20.40	13.61	29.76	28.43
	Pair packing	38.33	50.09	49.47	45.11	24.25	19.46	39.61	37.79
	BGE chunks	40.65	48.49	47.33	48.11	28.50	20.86	41.31	38.99
	SENTPACK (BGE)	41.79	49.78	56.18	49.58	31.84	20.85	45.87	41.67
	SENTPACK (hybrid)	41.28	52.99	51.86	49.42	30.37	19.95	43.89	40.98
Qwen2.5-14B	Full context	45.46	53.86	62.00	58.52	37.21	28.50	52.58	47.59
	LLMLingua-2	30.93	32.71	38.34	34.15	18.34	8.12	30.27	27.10
	Pair packing	39.37	49.88	56.88	52.25	31.82	19.92	46.98	41.69
	BGE chunks	39.75	49.05	55.42	53.34	35.28	21.19	48.02	42.34
	SENTPACK (BGE)	41.29	50.37	60.59	60.29	37.33	21.18	52.74	45.18
	SENTPACK (hybrid)	40.70	53.03	58.78	55.52	36.65	21.24	50.32	44.32

Table 9: Per-task F1 at 1,024 tokens (full context: up to 30K tokens).

Model	Comparison	512	1,024	2,048
Qwen2.5-7B	SENTPACK − BGE chunks	+3.69 [1.69, 5.68]	+1.79 [−0.08, 3.65]	+2.62 [0.82, 4.49]
	SENTPACK − Pair packing	+2.86 [1.09, 4.59]	+1.53 [−0.14, 3.17]	+2.92 [1.26, 4.58]
	SENTPACK-BGE − BGE chunks	+2.59 [0.91, 4.29]	+1.91 [0.21, 3.64]	+2.17 [0.43, 3.96]
Llama-3.1-8B	SENTPACK − BGE chunks	+3.17 [1.14, 5.17]	+1.99 [0.20, 3.84]	+1.93 [0.08, 3.80]
	SENTPACK − Pair packing	+2.39 [0.69, 4.07]	+3.19 [1.48, 4.89]	+3.18 [1.53, 4.80]
	SENTPACK-BGE − BGE chunks	+2.89 [1.24, 4.56]	+2.68 [1.06, 4.38]	+2.73 [1.03, 4.44]
Qwen2.5-14B	SENTPACK − BGE chunks	+4.17 [2.27, 6.12]	+1.98 [0.20, 3.85]	+2.21 [0.45, 3.99]
	SENTPACK − Pair packing	+3.71 [2.15, 5.26]	+2.63 [1.17, 4.12]	+3.27 [1.65, 4.90]
	SENTPACK-BGE − BGE chunks	+2.95 [1.38, 4.53]	+2.84 [1.25, 4.49]	+2.81 [1.12, 4.59]

Table 10: Per-model paired bootstrap differences (F1, 95% intervals). SENTPACK denotes the frozen hybrid configuration.

chunk packing by 2.1 and 1.9 F1 (two-model averages, all $p < 0.001$). Since SENTPACK outperforms the pair packer in every setting of Table 2, these baselines are dominated by SENTPACK as well; we report them here rather than in the main table because they were run in a different environment.